Welcome to the Formula 1 Database Challenge! In this activity, you'll be working with a comprehensive dataset containing information about Formula 1 racing, including data on constructors, drivers, races, and results from various seasons.

Download the dataset made up of 8 MS Excel files here before you start : constructor_standings.csv ; constructors.csv ; driver_standings.csv ;drivers.csv ;pitstops.csv ;results.csv ;results_history.csv ;results_qualy.csv

Your task is to navigate through this rich dataset using SQL queries, demonstrating your ability to extract, manipulate, and analyse data effectively. The challenges range from basic data retrieval to more complex operations involving joins, subqueries, and aggregate functions.

This activity is designed to test and enhance your SQL skills across various difficulty levels. You'll start with simple SELECT statements and progress to more advanced queries involving multiple tables and sophisticated data analysis techniques.

Each task is designed to simulate real-world scenarios that a data analyst might encounter when working with sports statistics or any large, relational database. As you work through these challenges, think about how each query could be applied in a professional setting to derive meaningful insights from the data.

Remember, the key to success in this activity is not just getting the correct results, but also understanding the logic behind each query and how it interacts with the database structure. Good luck, and may your queries run quickly and your insights be plentiful!

Are you ready to begin the Formula 1 Database Challenge?

1. List all Constructors:

- Retrieve the names of all constructors.

2. Find All Drivers:

- List the names of all drivers in the dataset.

3. Count the Number of Constructors:

- How many constructors are there in the dataset?

4. Count the Number of Drivers:

- How many drivers are there in the dataset?

5. List All Races for a Specific Season:

- Retrieve all races that took place in the 2020 season.

6. Basic SELECT:

- Retrieve the `Name` and `Nationality` of all constructors.

7. SELECT with Aliases:

- Retrieve driver `GivenName` and `FamilyName` and alias them as `FirstName` and `LastName`.

8. SELECT Distinct Values:

- Retrieve distinct nationalities of drivers from the `drivers` table.

9. SELECT with Calculated Columns:

- Retrieve the `Position` and `Points` of drivers, and calculate their `PointsPerPosition` (Points divided by Position).

10. SELECT with Concatenation:

- Retrieve driver `GivenName` and `FamilyName` concatenated into a single column named `FullName`.

11. Basic Filtering:

- Retrieve all races where the `Season` is 2022.

12. Filtering with Multiple Conditions:

- Retrieve drivers who are either `German` or `British`.

13. Filtering with LIKE:

- Retrieve all constructors whose name contains 'Ferrari'.

14. Filtering with IN:

- Retrieve all results where the `ConstructorID` is either 'ferrari' or 'williams'.

15. Filtering with Date:

- Retrieve all drivers who were born before 2000.

16. Basic Sorting:

- Retrieve all races sorted by `Season` in ascending order.

17. Sorting with Multiple Columns:

- Retrieve all drivers sorted first by `Nationality` and then by `GivenName`.

18. Descending Order:

- Retrieve all results sorted by `Points` in descending order.

19. Sorting with NULLs:

- Retrieve all drivers and sort by `DateOfBirth`, placing NULL values last.

20. Top N Results:

- Retrieve the top 5 drivers with the highest `Points` in the 2020 season.

21. Basic LIMIT:

- Retrieve the first 10 constructors from the `constructors` table.

22. LIMIT with OFFSET:

- Retrieve 10 drivers starting from the 11th driver in the list.

23. Top N Results with OFFSET:

- Retrieve the next 10 drivers after the top 5 drivers with the most `Points` in the 2021 season.

24. LIMIT without ORDER BY:

- Retrieve the first 5 results of the 2020 season without specifying the order.

25. Pagination:

- Retrieve drivers for the 2020 season, showing results 11 through 20.

26. SUM Function:

- Calculate the total `Points` scored in the 2024 season.

27. AVG Function:

- Calculate the average `Points` scored by drivers in the 2000 season.

28. MAX and MIN Functions:

- Find the maximum and minimum `Points` scored by a driver in the 2021 season.

29. COUNT Function:

- Count the number of races in the 2000 season.

30. GROUP_CONCAT Function:

- List all drivers in each constructor, concatenated into a single column.

31. Basic GROUP BY:

- Retrieve the total `Points` scored by each constructor in the 2000 season.

32. GROUP BY with HAVING:

- Retrieve constructors that have more than 20 `Points` in the 2002 season.

33. COUNT with GROUP BY:

- Count the number of races each constructor participated in during 2020.

34. SUM with GROUP BY:

- Calculate the total `Points` for each driver, grouped by `Nationality`.

35. GROUP BY with Multiple Columns:

- Retrieve the average `Points` for each constructor and season combination.

36. Inner Join:

- Retrieve driver names and their corresponding constructor names for races in the 2000 season.

37. Left Join:

- Retrieve all constructors and any drivers who raced for theM(include constructors with no drivers).

38. Right Join:

- Retrieve all results and the corresponding drivers for each result, including results with no drivers.

39. Left Join:

- Retrieve a list of all drivers and their corresponding results, including drivers who have not participated in any races.

40. Join with Multiple Tables:

- Retrieve the GivenName, FamilyName, and ConstructorName for each driver, along with their total Points earned in the 2000 season.

41. Simple Subquery:

- Retrieve drivers who have more points than the driver with the least points in the 2000 season.

42. Subquery in SELECT:

- Retrieve the `GivenName` and `FamilyName` of drivers along with their highest `Points` in any race.

43. Correlated Subquery:

- Retrieve constructors and their drivers where the driver’s `Points` is greater than the average `Points` for that constructor.

44. Simple CTE

- Retrieve the average points scored per driver in the 2000 season. Use a CTE to first calculate the total points scored by each driver, and then calculate the average from this aggregated data.

45. Complex CTE

- Retrieve drivers who have finished in the top 3 positions in more than 5 races in the 2000 season, along with their average points per race.